

Dynamic Stroke-Aware Hybrid Token Transformer (DSHTT) for Tamil Handwritten Numeral Recognition

Key Novel Insight

Tamil numerals are **stroke-dominant and structurally unique**, unlike MNIST digits.

Instead of treating them as plain images, we:

- Extract **stroke-level structural priors**
- Combine them with **lightweight token-based learning**
- Use **dynamic gating to fuse both representations**

This hybrid idea is **rare in literature**, especially for low-resource scripts.

Proposed Architecture

Stage 1: Stroke-Aware Preprocessing (Novel Component)

- Convert input image → **skeletonized representation**
- Extract:
 - stroke endpoints
 - curvature maps
 - direction vectors

Output = **Stroke Descriptor Map (SDM)**

✓ Very lightweight (no heavy computation)

✓ Adds structural intelligence

Stage 2: Dual-Branch Lightweight Encoder

Branch A: Tiny CNN (Spatial Features)

- 3–4 depthwise separable conv layers
- Inspired by MobileNetV3
- Captures local texture

Branch B: Patch Token Encoder (Transformer-style)

- Split image into patches (like ViT, but tiny)
- Pass through:

- 2–3 lightweight attention blocks
- Inspired by Vision Transformer but compressed

✓ Use **linear attention** (reduces complexity)

✓ Keeps parameter count low

Stage 3: Dynamic Feature Gating (Core Novelty)

Instead of simple concatenation:

Introduce: **Stroke-Guided Dynamic Gating Unit (SDGU)**

- Input:
 - CNN features
 - Transformer tokens
 - Stroke Descriptor Map
- Learns:
 - Which branch to trust more for each sample

✓ This is **highly novel and patentable**

✓ Makes model adaptive to writing styles

Stage 4: Prototype-Based Classification (Efficiency Boost)

Instead of dense FC layers:

- Learn **class prototypes** (embedding centers)
- Use:
 - cosine similarity for classification

✓ Reduces parameters

✓ Improves generalization for small datasets